

Taha Efe Karabacak

Electrical and Electronics Engineering Student

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ABOUT ME

I am an Electrical and Electronics Engineering student at Karabük University. I aim to become a versatile engineer by combining electronics and engineering discipline with software and artificial intelligence. I develop projects on AI-powered applications, automation systems and embedded systems using Python, C#, SQL and other technologies.

EDUCATION AND TRAINING

SEP 2025 – JUN 2030 – KARABÜK, TÜRKIYE

Electrical and Electronics Engineering (Bachelor's)

Karabük University · language of instruction: 100% English

GRADUATED 2024 – KARABÜK, TÜRKIYE

High School Diploma

Vakıfbank Zübeyde Hanım Anatolian High School

WORK EXPERIENCE

SEP 2026 – PRESENT – KARABÜK, TÜRKIYE

Electronics Unit Member

Deep Dive Dynamics

- Extensive research and market analysis on electronic hardware and components.
- Supporting the engineering team in system design, hardware integration and testing during TEKNOFEST preparations.
- Developed a magnetic relay-based protection and circuit-interruption solution; designed an electromechanical protection architecture aimed at safely interrupting the circuit under fault conditions and ensuring system safety.
- Designed, in KiCad, the schematic of an XL4015-based step-down (buck) regulator that produces 5 V (about 5 A) from a 10–24 V input, for a Jetson load.

KARABÜK, TÜRKIYE

Private Tutor (individual)

Mathematics, Science and English · students aged 6–14

LANGUAGE SKILLS

MOTHER TONGUE: TURKISH

OTHER LANGUAGE: ENGLISH (EF SET, 26.08.2026 · 89/100)

Listening	**C2**
Reading	**C2**
Speaking	**C2**
Writing	**C2**

DIGITAL AND TECHNICAL SKILLS

Python | C# | .NET | C / C++ | Go | SQL | JavaScript | TypeScript | OOP

HTML5 / CSS3 | React | Next.js | Vite | FastAPI | PyQt5

Proteus | KiCad | Altium Designer (learning) | Arduino | Tinkercad | Groq API

Whisper | Ollama

Taha Efe Karabacak

BTK Akademi certificates can be verified at btkakademi.gov.tr using the certificate number. All links are available at tahaefekarabacak.com.tr/blog.

CERTIFICATES AND TRAINING

14.10.2026	**Occupational Health and Safety Basic Training** · Karabük University OHS Coordination Office
07.10.2026	**Learning SQL with Applications** · BTK Akademi
06.10.2026	**Software Engineer** · HackerRank
06.10.2026	**C# (Basic)** · HackerRank
06.10.2026	**Go (Intermediate)** · HackerRank
01.10.2026	**Introduction to Firewalls** · BTK Akademi
30.09.2026	**Anthropic Claude** · BTK Akademi
26.08.2026	**Problem-Solving Techniques** · BTK Akademi
26.08.2026	**EF SET English (C2)** · EF SET
16.07.2026	**ChatGPT** · BTK Akademi
16.07.2026	**Gemini Veo** · BTK Akademi

PROJECTS

J.A.R.V.I.S. AI Assistant · 2025 – present
Desktop assistant built with PyQt5 and controlled by voice. Speech via Groq API (Llama 3.3-70B), Whisper and Edge-TTS; persistent memory, autonomous screen control and a self-update module.

J.A.R.V.I.S. intelligence dashboard
Dashboard built with Next.js 14, TailwindCSS, FastAPI, MongoDB and a local LLM (Ollama), showing news, earthquake, flight, weather and finance data.

Finance Control · 2026
Finance dashboard tracking currencies, crypto and commodities with CBRT and CoinGecko data (React, TypeScript, Vite).

Student Sharing Network · 2026
Platform for sharing lecture notes and past exam questions (React, TypeScript, Vite).

Smart temperature monitoring and alarm system
Arduino-based design with DHT11, LEDs and a buzzer (Tinkercad).

NFC / QR table menu cards
Card designs with Google review, Instagram, menu and Wi-Fi QR codes for businesses in Safranbolu.

Personal website and blog · 2026
tahaefekarabacak.com.tr/blog

INTERESTS

Artificial intelligence | Embedded systems | Cybersecurity
Chess (tournament experience) | Music | Drawing | Sports